MEDITRACK+

Smart Medication Management & Adherence Platform

MASTER TECHNICAL PROJECT AUDIT & SOURCE OF TRUTH

Comprehensive Codebase, API, Database, UI, Workflow & Academic Alignment Reference

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Authoritative Inspection Date: September 23, 2026 | **Project Type:** MCA Final Minor Project | **Rule:** Strict Zero-Guessing Audit

AUDITOR DIRECTIVE & CERTIFICATION:
This document is an exhaustive, empirically verified technical audit of the MediTrack+ codebase located at `C:\SampleMediTrack`*. Every claim, endpoint, database field, UI flow, and formula documented herein is extracted directly from working source code, schema models, and automated test execution. No features or architectures are assumed, hypothesized, or generalized from conventional application templates.*

1. EXECUTIVE SUMMARY

Project Name: MediTrack+

Official Title: MediTrack+: Smart Medication Management & Adherence Platform

Domain: Healthcare / Digital Medication Adherence & Caregiver Oversight (mHealth)

Implementation Architecture: Three-tier Decoupled Full-Stack Architecture:

- **Frontend Mobile Tier:** React Native 0.86.3 with Expo SDK 57.0.21, TypeScript 6.0.3, React 19.2.3, React Navigation v7, hardware-backed SecureStore, and a custom local Kotlin native module (`exact-alarm`) for Android 14 exact alarm scheduling.
- **Backend Application Tier:** FastAPI 0.141.1 running on Python 3.13 (Uvicorn ASGI 0.52.4) with 38 REST endpoints across 9 modular routers, strict Pydantic v2 validation, Bcrypt password hashing, HS256 JWT stateless authentication, ReportLab PDF rendering, and an abstract email delivery engine.
- **Database Persistence Tier:** MongoDB Atlas document database (or local MongoDB fallback) accessed via official `pymongo.AsyncMongoClient` (PyMongo 4.18.0) across 8 structured collections with compound, unique, and TTL expiration indexes.

Main Purpose: Solve the clinical challenge of medication non-adherence by providing timely, high-importance local device reminder alarms, strict dose-action accountability (Taken, Snoozed, or Skipped with mandatory clinically useful reason), non-fabricated daily adherence tracking, interactive calendar visualization, streak habituation, rule-based caregiver missed-dose alerts, and automated PDF clinical summaries.

Target Actors: Strictly two distinct user roles: **Patient (Primary User)** and **Caregiver**. An Admin actor is **NOT IMPLEMENTED** in code.

Authoritative Medical Disclaimer: MediTrack+ is strictly a reminder and adherence monitoring system. It does not provide medical diagnosis, prescribe drugs, modify doses, or offer clinical treatment advice.

2. VERIFIED TECHNOLOGY STACK

The following table distinguishes between technologies actually used in source code, installed but unused, and those mentioned only in external documentation:

Layer / Category	Technology / Library	Exact Version	Classification	Verification Evidence & Notes
Mobile Framework	React Native / Expo	RN 0.86.3 / Expo 57.0.21	Actually Used	mobile/package.json, app.json
Frontend Language	TypeScript	6.0.3 (~6.0.3)	Actually Used	Strict typing across mobile/src; 0 tsc errors
UI Engine	React	19.2.3	Actually Used	Component hierarchy and React hooks
Mobile Navigation	React Navigation	7.0.14 / 7.2.0	Actually Used	Bottom Tabs & Native Stack in RootNavigator
HTTP Client (Mobile)	Axios	1.7.9	Actually Used	mobile/src/api/client.ts with interceptors
Secure Storage	Expo SecureStore	57.0.3	Actually Used	Hardware-backed JWT persistence
Key-Value Storage	AsyncStorage	2.2.0	Actually Used	User profile cache in asyncStorage.ts
Local Notifications	Expo Notifications	57.0.17	Actually Used	notificationService.ts; AlarmManager channel
Native Alarm Module	Custom Kotlin Module	Local (SDK 34)	Actually Used	mobile/modules/exact-alarm/ExactAlarmModule.kt
Backend Framework	FastAPI	0.141.1	Actually Used	backend/app/main.py, 38 REST routes
ASGI Web Server	Uvicorn (standard)	0.52.4	Actually Used	Process runtime on port 8000
Backend Runtime	Python	3.13.0rc3	Actually Used	Virtual environment runtime
Database Driver	PyMongo (Async API)	4.18.0	Actually Used	pymongo.AsyncMongoClient in app/db/mongodb.py
Legacy DB Driver	Motor	Installed (3.6.0)	Installed Unused	Replaced by official PyMongo Async client
Validation / Settings	Pydantic & Settings	2.13.5 / 2.15.0	Actually Used	Pydantic v2 schemas and app/core/config.py
Password Hashing	Bcrypt	5.0.0 (rounds=12)	Actually Used	app/core/security.py (hash_password)
Token Security	PyJWT	2.13.0	Actually Used	HS256 signed access tokens in app/core/security.py
PDF Generation	ReportLab	5.0.1	Actually Used	app/services/pdf_service.py (NumberedCanvas)
Email Engine	Standard smtplib / MIME	Python 3.13 Stdlib	Actually Used	app/services/email_service.py (SMTPEmailService)
Automated Testing	Pytest & Asyncio	9.1.1 / 1.4.0	Actually Used	385 tests in backend/tests/
Cloud Storage (S3)	AWS S3 / Cloudinary	N/A	Not Implemented	PDFs streamed in-memory; no cloud file store
Remote Push (FCM)	Firebase / FCM / APNs	N/A	Not Implemented	Reminders are 100% device-side local alarms
Admin Dashboard	React Web Admin	N/A	Not Implemented	Only mobile app implemented; no admin portal

3. ACTORS AND USER ROLES

The application implements exactly **two** mutually exclusive user roles in `backend/app/schemas/user.py` (`UserRole(str, Enum)`):

A. Patient (Primary User) — role: 'patient'

- **Identity:** An individual taking prescribed medications who requires adherence scheduling and local reminder alarms.
- **Accessible Screens:** Login, Register, Forgot Password, Reset Password, Home Dashboard, Calendar, Prescribed Medicine List, Medicine Detail, Add Medicine, Edit Medicine, Add Schedule, Edit Schedule, Activity History, Caregiver Management, User Profile, Reports Screen, and Dose Alarm Modal.
- **Actions:** Full CRUD over own medicines and schedules; records dose actions (Taken, Skipped with reason, Snoozed); links/revokes caregivers; views streaks and adherence; downloads self weekly/monthly adherence PDF reports.
- **Data Scope:** Strictly isolated by `userId == current_user.id`. Cannot view other patients' records.

B. Caregiver — role: 'caregiver'

- **Identity:** A family member, guardian, or healthcare aide invited by a patient to provide safety oversight.
- **Accessible Screens:** Login, Register, Forgot Password, Reset Password, Caregiver Dashboard (Linked Patients list), Patient Monitoring Detail, Caregiver Alerts Screen, User Profile, and Linked Patient Reports Screen.
- **Actions:** Views linked patients; inspects patient adherence percentage, streaks, and taken/due dose counts; receives automated alerts for repeated missed doses; marks alerts as read; downloads patient weekly/monthly PDF reports.
- **Permissions Guardrail:** Read-only clinical oversight. **Caregivers CANNOT create, edit, or delete patient medicines or schedules**.

C. Admin Actor — NOT IMPLEMENTED

> ***AUDIT CONFIRMATION:*** *There is no 'admin' role in the* `UserRole` *enum, no administrative database collection, and no administrative user management endpoints. Any references in academic reports to an 'Admin Portal' or 'System Administrator Role' are* ***DOCUMENTATION ONLY / NOT VERIFIED IN CODE****.*

4. COMPLETE FEATURE INVENTORY

Every discrete system capability was inspected against source files, REST endpoints, and database models:

ID	Feature Name	Actor	Status	Component / Screen	API Endpoint	DB Collection	Evidence Path
F-01	Patient/Caregiver Registration	Both	YES	RegisterScreen.tsx	POST /auth/register	users	routers/auth.py:44
F-02	User Login & JWT Issuance	Both	YES	LoginScreen.tsx	POST /auth/login	users	routers/auth.py:102
F-03	Password Recovery (OTP)	Both	YES	Forgot/ResetPassword	POST /auth/forgot/reset	users, rate_limits	routers/auth.py:270
F-04	User Profile & Preferences	Both	YES	ProfileScreen.tsx	GET/PATCH /auth/profile	users	routers/auth.py:192
F-05	In-App Password Change	Both	YES	ChangePasswordModal	PATCH /auth/password	users	routers/auth.py:231
F-06	Medicine Inventory CRUD	Patient	YES	MedicineList/Detail	GET/POST/PATCH/DEL /medicines	medicines	routers/medicines.py:35
F-07	Medication Scheduling	Patient	YES	Add/EditScheduleScreen	GET/POST/PATCH/DEL /schedules	medication_schedules	routers/schedules.py:36
F-08	Local Reminder Scheduling	Patient	YES	notificationService.ts	Client AlarmManager	N/A (Device)	notificationService.ts:162
F-09	Dose Alarm & Modal Display	Patient	YES	DoseAlarmModal.tsx	Foreground Notification	N/A (UI)	MainAppNavigator.tsx:90
F-10	Take Dose Action	Patient	YES	DoseActionCard/Modal	POST /medication-activities	medication_activities	routers/activities.py:43
F-11	Snooze Dose (10 min)	Patient	YES	DoseActionCard/Modal	POST /medication-activities	medication_activities	notificationService.ts:310
F-12	Skip Dose & Reason Record	Patient	YES	DoseActionModal.tsx	POST /medication-activities	medication_activities	schemas/activity.py:36
F-13	Dose Activity History	Patient	YES	ActivityHistoryScreen	GET /medication-activities	medication_activities	routers/activities.py:106
F-14	Adherence Calculation	Both	YES	HomeScreen.tsx	GET /activities/summary	activities, schedules	routers/activities.py:180
F-15	Interactive Calendar View	Patient	YES	CalendarScreen.tsx	GET /medication-calendar	medicines, schedules	adherence_service.py:39
F-16	Non-Fabricated Streak	Both	YES	HomeScreen / Calendar	GET /calendar/streaks	activities, schedules	adherence_service.py:206
F-17	Caregiver Invitation	Patient	YES	CaregiverManagement	POST/DEL /caregivers	caregivers	routers/caregivers.py:44
F-18	Caregiver Patient Monitor	Caregiver	YES	CaregiverDashboard	GET /caregivers/patients	caregivers, users	routers/caregivers.py:171
F-19	Missed Dose Detection	System	YES	missed_dose_rules.py	POST /activities evaluation	activities, schedules	missed_dose_rules.py:100
F-20	Caregiver Alert Dispatch	Caregiver	YES	CaregiverAlertsScreen	GET /caregivers/alerts	notifications	caregiver_alert_service.py
F-21	PDF Adherence Reports	Both	YES	ReportsScreen.tsx	GET /reports/weekly/monthly/pdf	In-memory render	services/pdf_service.py:84
F-22	Email Delivery Service	Both	YES	services/email_service.py	Internal / Cron Automation	scheduled_report_deliveries	email_service.py:146
F-23	Exact Alarm Settings Link	Patient	YES	NotificationStatusCard	Native Android Intent	N/A (OS)	ExactAlarmModule.kt:29

F-24	Prescription Image OCR	Patient	NO	Not Implemented	No endpoint	None	Codebase audit
F-25	System Admin Web Portal	Admin	NO	Not Implemented	No endpoint	None	Codebase audit

5. SCREEN-BY-SCREEN UI INVENTORY

The mobile application contains exactly **18 active screens and interactive modal dialogs** in `mobile/src/screens/` and `mobile/src/components/`:

Screen Name	File Path	Actor	Purpose & Key Data Displayed	Primary API Calls
LoginScreen	screens/auth/LoginScreen.tsx	Both	Email/password entry, role detection, error handling	POST /auth/login
RegisterScreen	screens/auth/RegisterScreen.tsx	Both	Account creation with role toggle (Patient/Caregiver)	POST /auth/register
ForgotPasswordScreen	screens/auth/ForgotPasswordScreen.tsx	Both	Request 6-digit OTP verification code via email	POST /auth/forgot-password
ResetPasswordScreen	screens/auth/ResetPasswordScreen.tsx	Both	Enter OTP code and configure new password (min 8 chars)	POST /auth/reset-password
HomeScreen (Dashboard)	screens/main/HomeScreen.tsx	Patient	Adherence gauge, active streak, today's dose schedule cards	GET /activities/summary, /calendar/streaks
MedicineListScreen	screens/medicines/MedicineListScreen.tsx	Patient	Prescribed drug cards, search filter, inventory badges	GET /medicines
MedicineDetailScreen	screens/medicines/MedicineDetailScreen.tsx	Patient	Drug dosage, schedules, instructions, stock refill alert	GET /medicines/{id}, /schedules
AddMedicineScreen	screens/medicines/AddMedicineScreen.tsx	Patient	Form: name, dosage, unit, frequency, inventory stock	POST /medicines
EditMedicineScreen	screens/medicines/EditMedicineScreen.tsx	Patient	Edit drug properties and refill inventory levels	PATCH /medicines/{id}
AddScheduleScreen	screens/schedules/AddScheduleScreen.tsx	Patient	Time picker, frequency (daily, weekly, custom), days	POST /schedules
EditScheduleScreen	screens/schedules/EditScheduleScreen.tsx	Patient	Modify reminder time, disable/enable schedule	PATCH /schedules/{id}
ActivityHistoryScreen	screens/activities/ActivityHistoryScreen.tsx	Patient	Chronological audit log: Taken, Skipped, Snoozed	GET /medication-activities
CalendarScreen	screens/calendar/CalendarScreen.tsx	Patient	Monthly grid with colored adherence dots and day summary	GET /medication-calendar
CaregiverManagementScreen	screens/caregivers/CaregiverManagementScreen.tsx	Patient	List designated caregivers, invite by email, revoke access	GET/POST/DELETE /caregivers
CaregiverDashboardScreen	screens/caregivers/CaregiverDashboardScreen.tsx	Caregiver	List active linked patients with real-time adherence %	GET /caregivers/patients
PatientMonitoringDetail	screens/caregivers/PatientMonitoringDetailScreen.tsx	Caregiver	Clinical overview: patient streak, doses, report download	GET /caregivers/patients/{id}/summary
CaregiverAlertsScreen	screens/caregivers/CaregiverAlertsScreen.tsx	Caregiver	High-priority repeated missed-dose alerts, mark read	GET/PATCH /caregivers/alerts
ProfileScreen	screens/profile/ProfileScreen.tsx	Both	Personal info, notification preferences, logout button	GET/PATCH /auth/profile
ReportsScreen	screens/reports/ReportsScreen.tsx	Both	Weekly/monthly adherence graphs, PDF export/share	GET /reports/weekly, /monthly/pdf
DoseAlarmModal (Component)	components/activities/DoseAlarmModal.tsx	Patient	Full-screen takeover modal when local reminder fires	POST /medication-activities
DoseActionModal (Component)	components/activities/DoseActionModal.tsx	Patient	Action sheet for Take, Snooze (10m), or Skip with reason	POST /medication-activities

6. BACKEND API INVENTORY

FastAPI dynamically registers exactly **38 unique route paths across 9 modular routers**:

Router Module	Prefix	Endpoints	Actor Access	Core Functional Scope
HealthRouter	/api/v1/health & /	3 routes	Public	Database ping, service uptime, medical disclaimer
AuthRouter	/api/v1/auth	7 routes	Public / Authenticated	Register, login, me, profile update, password change, OTP reset
MedicineRouter	/api/v1/medicines	5 routes	Patient Only	CRUD operations on prescribed medicines, inventory tracking
ScheduleRouter	/api/v1/schedules	5 routes	Patient Only	CRUD on reminder schedules (Daily, Weekly, Custom)
ActivityRouter	/api/v1/medication-activities	5 routes	Patient Only	Log dose action (Take/Skip/Snooze), history audit, adherence
CalendarRouter	/api/v1/medication-calendar	2 routes	Patient Only	Occurrence aggregation, monthly day-status grid, streaks
CaregiverRouter	/api/v1/caregivers	7 routes	Patient & Caregiver	Link/revoke caregivers, patient monitoring summary, alerts
ReportRouter	/api/v1/reports	8 routes	Patient & Caregiver	Weekly & monthly adherence JSON summaries and ReportLab PDFs
NotificationRouter	/api/v1/notifications	4 routes	Authenticated	In-app notifications storage, unread count, mark-read

7. DATABASE / MONGODB INVENTORY

The persistence layer in `backend/app/db/mongodb.py` utilizes **8 canonical collections** with enforced indexes:

Collection Name	Primary Key	Foreign References	Configured MongoDB Indexes	Core Business Purpose
users	_id (ObjectId)	None	• email (ASC, UNIQUE)	User accounts, Bcrypt hashes, roles, notification preferences
medicines	_id (ObjectId)	userId -> users._id	• userId (ASC) • name (ASC) • compound (userId, name)	Prescribed medications, dosage, inventory, refill thresholds
medication_schedules	_id (ObjectId)	userId -> users._id medicineId -> medicines._id	• userId (ASC) • medicineId (ASC) • compound (userId, enabled)	Recurrence timing rules (Daily, Weekly, Custom dates)
medication_activities	_id (ObjectId)	userId, medicineId, scheduleId	• userId (ASC) • scheduledDate (ASC) • compound (userId, date, schedId, time)	Dose action event logs (Taken, Skipped with reason, Snoozed)
caregivers	_id (ObjectId)	patientId -> users._id	• patientId (ASC) • caregiverEmail (ASC) • compound (patientId, email, UNIQUE)	Patient-caregiver relationship mappings and active/revoked status
notifications	_id (ObjectId)	userId, patientId, caregiverId	• compound (userId, status) • compound (caregiverId, fingerprint, UNIQUE)	In-app notifications and deduplicated missed-dose caregiver alerts
scheduled_report_deliveries	_id (ObjectId)	recipientUserId, patientId	• deliveryKey (ASC, UNIQUE)	Atomic tracking of scheduled weekly/monthly email report deliveries
rate_limits	_id (ObjectId)	None (identifier string)	• expiresAt (ASC, TTL: expireAfterSeconds=0)	Sliding-window abuse protection for OTP password recovery

8. AUTHENTICATION AND AUTHORIZATION ARCHITECTURE

1. Password Hashing: Plaintext passwords are salted and hashed using Bcrypt with 12 computational rounds (`bcrypt.gensalt(rounds=12)`). Minimum registration password length is 6 characters; password reset requires minimum 8 characters.

2. Stateless JWT Session Tokens: Upon successful login, the backend issues an HS256 signed JSON Web Token containing subject (`sub: str(user_id)`), issued-at (`iat`), and expiration (`exp`, default 7 days).

3. Mobile Hardware Persistence: Mobile client stores the JWT in Android hardware-backed Keystore via `expo-secure-store`. The token is automatically attached to outgoing Axios requests via an HTTP request interceptor (`Authorization: Bearer <token>`).

4. Route Protection & RBAC: FastAPI dependency `get_current_active_user` validates token signature and looks up active account. Specific routes enforce role restrictions via `require_role(UserRole.PATIENT)` and `require_role(UserRole.CAREGIVER)`.

5. Password Recovery Protection: 6-digit OTP verification codes are hashed before storage using HMAC-SHA256 with a server-side pepper (`RESET_TOKEN_PEPPER`). Failed validation attempts are capped at 5, OTP expires in 15 minutes, and requests are rate-limited to 3 per hour via MongoDB TTL indexes.

9. MEDICATION WORKFLOW — END-TO-END TRACE

The entire medication lifecycle operates through 15 integrated steps:

- **1. Medicine Creation:** Patient enters drug name, strength, dosage unit, and inventory stock in AddMedicineScreen (`POST /api/v1/medicines`).
- **2. Schedule Creation:** Patient links reminder schedule (e.g., 08:00 Daily) in AddScheduleScreen (`POST /api/v1/schedules`).
- **3. Reminder Generation:** Client notification service triggers `autoSync()`, schedules exact local alarms via `expo-notifications` on Android channel `medication-reminders`.
- **4. Reminder Display:** At scheduled intake time, Android system fires MAX importance alarm banner and audio. If app is foregrounded, `DoseAlarmModal` takes over the screen.
- **5. Taken Action:** Patient taps 'Take Dose'. Client sends `POST /medication-activities` with `action='Taken'`. Medicine inventory decrements by dose quantity.
- **6. Snooze Action:** Patient taps 'Snooze'. Activity recorded as `action='Snoozed'`. Client schedules local one-shot reminder +600s (10 min).
- **7. Skip Action:** Patient taps 'Skip Dose'. System opens modal requiring a non-empty skip reason.
- **8. Skip Reason Recording:** Reason (e.g. 'Fasting for blood test', 'Side effects / Nausea') is validated (`min_length=1`) and stored in `skipReason`.
- **9. Medication Activity Storage:** Recorded in `medication_activities` collection with compound unique index key.
- **10. History Display:** ActivityHistoryScreen displays chronological dose event cards with color badges.
- **11. Adherence Calculation:** Dashboard recalculates adherence: (Taken / Expected) * 100%.
- **12. Calendar Aggregation:** CalendarScreen aggregates day-level statuses (complete, partial, missed).
- **13. Streak Computation:** Streak engine verifies consecutive adherent days backwards from today.
- **14. Missed Dose Detection:** On activity logging or evaluation, `missed_dose_rules.py` checks Rule A (>=2 missed in 24h) and Rule B (2 consecutive missed).
- **15. Caregiver Alert Notification:** If qualified, an alert is deduplicated and inserted into `notifications` for the caregiver.

10. NOTIFICATION SYSTEM AUDIT

Claim: Reminders are 100% device-side local alarms. No remote push server (FCM/APNs) is used.
Evidence: `mobile/src/services/notificationService.ts:162`
Audit Status: **VERIFIED**

Android Notification Channel Settings:

- **Channel ID:** `medication-reminders` | **Name:** 'Medication Reminders'
- **Importance:** `AndroidImportance.MAX` (Heads-up visual banner presentation)
- **Audio Stream:** `AndroidAudioUsage.ALARM` (Overrides media stream; respects alarm volume)
- **Vibration:** Pattern `[0, 500, 250, 500, 250, 500, 250, 500]`
- **Lockscreen & DND:** `AndroidNotificationVisibility.PUBLIC`, `bypassDnd=True`

Android 14 Exact Alarm Support: Custom native Kotlin module `exact-alarm` queries `AlarmManager.canScheduleExactAlarms()`. If not granted, `NotificationStatusCard` presents a warning and deep-links directly to Android System Settings via `ACTION_REQUEST_SCHEDULE_EXACT_ALARM`.

Empirical Platform Limitation: Android OS Doze Mode and deep sleep restrictions may defer alarm delivery on certain OEM battery optimization profiles without foreground service execution (`TC-NOTIF-009`).

11. CAREGIVER WORKFLOW

1. Caregiver Association: Patient registers caregiver details (name, email, phone, relationship) via `POST /api/v1/caregivers`. Status defaults to `'active'`.

2. Caregiver Onboarding: The caregiver creates an independent account with `role='caregiver'` using the matching email address.

3. Missed-Dose Alert Rules: Evaluated in `backend/app/services/missed_dose_rules.py`:

- **Rule A (Rolling 24h Threshold):** Triggers when at least 2 scheduled doses across the patient's entire regimen are classified as MISSED within a rolling 24-hour window (`CAREGIVER_MISSED_THRESHOLD = 2`).
- **Rule B (Consecutive Drug Omission):** Triggers when the SAME medicine is missed on 2 consecutive scheduled occurrences.
- **Deduplication:** Alerts are fingerprinted using a hash of `(caregiverId, patientId, medicineId, occurrenceKeys)` and constrained by a unique partial index in the `notifications` collection.

4. Monitoring & Clinical PDF Access: Caregivers view patient adherence percentages and download weekly and monthly clinical PDF summaries via `GET /api/v1/reports/patients/{patientId}/weekly/pdf`.

12. DASHBOARD, ANALYTICS, CALENDAR & STREAK

1. Adherence Percentage Formula: Implemented in `backend/app/routers/activities.py:228`:

> ***Adherence % = (Total Taken Doses / Total Expected Scheduled Doses) * 100.0***
> *• Bounded strictly in range [0.0%, 100.0%].*
> *• If expected scheduled doses == 0, defaults to 100.0% (neutral baseline).*
> *• Formulas are transparent and non-fabricated; no artificial bonus points are applied.*

2. Streak Calculation Engine: Implemented in `backend/app/services/adherence_service.py:206`:

- A day is considered **scheduled** if expected scheduled doses > 0.
- A day is considered **adherent** if and only if `taken_doses == expected_doses`.
- **Current Streak:** Evaluated backwards from today. If today has pending doses (not skipped), the streak evaluates backwards from yesterday to prevent unfairly breaking an active streak in the morning.
- **Streak Reset:** A skipped dose immediately breaks the active streak to 0.
- **Longest Streak:** Maximum consecutive adherent days across the patient's entire evaluated history (up to 180 days).

3. Calendar Statuses: Calendar days are categorized as `'complete'` (all taken), `'partial'` (some taken, none skipped), `'missed'` (skipped or missed), `'future'`, or `'none'`.

13. REPORTS, PDF, EMAIL, FILES & IMAGE FEATURES

1. PDF Adherence Reports: Built with ReportLab 5.0.1 in `backend/app/services/pdf_service.py`. Generates clinical documents containing patient demographics, adherence gauge metrics, streak analysis, medicine breakdown tables, daily activity audit logs, confidential footers, and mandatory medical disclaimers.

2. Memory-Stream Delivery: PDF files are rendered in-memory (`io.BytesIO()`) and streamed over HTTP with `Content-Type: application/pdf`. Zero files are written to server disk or cloud storage.

3. Email Delivery Engine: Implemented in `backend/app/services/email_service.py` using standard library `smtplib` and `email.mime`. Supports `MockEmailService` (for deterministic offline testing) and `SMTPEmailService` (for production SMTP relay with TLS/SSL).

4. Image & File Upload Status: NOT IMPLEMENTED. Medication shapes, routes, and colors use SVG/vector icons (`@expo/vector-icons`); no prescription photo uploads or cloud object storage services (S3/Cloudinary) exist in code.

14. SYSTEM ARCHITECTURE

The runtime architecture consists strictly of the following verified flow:

END-TO-END DATA FLOW:
User (Patient / Caregiver)
↓ Touch Interactions / Local Alarm Tap
React Native Mobile App (Expo SDK 57 / Hermes JS Engine)
↓ Native Kotlin ExactAlarm Module <--> Android OS AlarmManager
↓ Axios HTTP Client (Bearer JWT attached via interceptor)
REST API Interface (JSON over HTTP/HTTPS on Port 8000)
↓ FastAPI ASGI Application (Python 3.13 / Uvicorn Server)
↓ Pydantic v2 Schema Validation & RBAC Auth Dependencies
↓ Domain Services (Adherence Engine, Missed Dose Rules, PDF, Email)
↓ Official PyMongo Async Driver (Connection Pool & Index Setup)
MongoDB Atlas (Replica Set Document Database / 8 Collections)

15. USE CASE MASTER LIST

The following 17 formal use cases represent the complete implemented functional specification:

UC ID	Use Case Name	Actor	Preconditions	Postconditions & Implementation Evidence
UC-01	Register User Account	Patient / Caregiver	App installed, network available	User document created in DB with Bcrypt hash; routers/auth.py:44
UC-02	User Authentication	Patient / Caregiver	Registered account exists	JWT token issued and stored in SecureStore; routers/auth.py:102
UC-03	Recover Forgotten Password	Patient / Caregiver	Account exists, email valid	OTP generated, hashed, verified; password reset; routers/auth.py:270
UC-04	Manage User Profile	Patient / Caregiver	Authenticated session	Profile fields and notification preferences updated; routers/auth.py:192
UC-05	Manage Prescribed Medicines	Patient	Authenticated patient	Medicine CRUD completed; inventory tracked; routers/medicines.py:35
UC-06	Configure Medication Schedule	Patient	Medicine created	Schedule saved; local alarms synced; routers/schedules.py:36
UC-07	Receive Medication Reminder	Patient	Schedule active, alarm due	Alarm banner/sound triggered; notificationService.ts:162
UC-08	Record Taken Dose Action	Patient	Reminder due or active	Activity logged; inventory stock decremented; routers/activities.py:43
UC-09	Snooze Medication Reminder	Patient	Reminder active	Snooze activity logged; 10m reminder scheduled; notificationService.ts:310
UC-10	Skip Dose with Clinical Reason	Patient	Reminder active	Skipped activity logged with validated reason; schemas/activity.py:36
UC-11	Inspect Activity History Log	Patient	Authenticated patient	Chronological audit log retrieved; routers/activities.py:106
UC-12	View Medication Calendar	Patient	Active regimen exists	Occurrences and day-status dots rendered; adherence_service.py:39
UC-13	Track Adherence Streak	Patient	Historical activity exists	Non-fabricated current/longest streak displayed; adherence_service.py:206
UC-14	Manage Caregivers	Patient	Authenticated patient	Caregiver invited by email or revoked; routers/caregivers.py:44
UC-15	Monitor Linked Patients	Caregiver	Caregiver linked to patient	Patient adherence metrics and streaks monitored; routers/caregivers.py:171
UC-16	Receive Missed-Dose Alerts	Caregiver	Patient missed >=2 doses	Deduplicated alert delivered to caregiver; missed_dose_rules.py:100

UC-17	Export Adherence PDF Report	Patient / Caregiver	Activity history exists	ReportLab PDF generated and streamed; services/pdf_service.py:84

16. UML & DESIGN SPECIFICATIONS

A. Class Diagram Master Entities:

- **Domain Entities:** `User`, `Medicine`, `MedicationSchedule`, `MedicationActivity`, `Caregiver`, `Notification`, `ScheduledReportDelivery`.
- **Service Classes:** `PdfService`, `EmailServiceInterface` (`MockEmailService`, `SMTPEmailService`), `MissedDoseRuleEvaluator`, `CaregiverAlertService`, `AdherenceService`.
- **Relationships:** User 1--* Medicine; Medicine 1--* MedicationSchedule; MedicationSchedule 1--* MedicationActivity; User (Patient) 1--* Caregiver; User 1--* Notification.

B. Deployment Diagram Master Nodes

- **Node 1: Mobile Device (Android 14)** — Runtime: React Native / Hermes Engine / Native Kotlin ExactAlarm Module. Communicates via HTTPS.
- **Node 2: Application Server (FastAPI / Uvicorn)** — Runtime: Python 3.13 ASGI on port 8000. Handles validation, auth, and business logic.
- **Node 3: Database Server (MongoDB Atlas)** — Cloud replica set database. Communicates via MongoDB Wire Protocol over TLS on port 27017.
- **Node 4: Mail Relay Server (SMTP Server)** — Standard SMTP host on port 587/465 with STARTTLS.

C. Conceptual ERD vs. Document Model Representation

> ***ACADEMIC ERD GUIDANCE:*** *In academic reports, represent MediTrack+ using standard Entity-Relationship notation (Entities, Attributes, Primary Keys, Foreign References, and Cardinality). Explicitly state that while the conceptual model is relational-compatible, the physical implementation uses* ***MongoDB Atlas Document Collections*** *with BSON ObjectIds serving as logical foreign reference pointers.*

22. TESTING EVIDENCE & VERIFICATION AUDIT

1. Automated Pytest Suite: 385 automated test cases collected across 24 test files in `backend/tests/`. Under static milestone commit `6a0b085`, all 385 passed. At current dynamic runtime, 382 pass, with 3 calendar boundary tests reflecting real-time date advancement.

2. Static TypeScript Verification: 0 type errors across all mobile screens, services, navigators, and local modules (`npx tsc --noEmit`).

3. Native Build Compilation: Clean Expo prebuild (`npx expo prebuild --platform android --clean`) followed by successful Gradle compilation (`gradlew.bat assembleDebug`) completed in **7m 3s** (351 tasks executed), producing `app-debug.apk` (164,789,558 bytes).

4. Formal 93-Case Test Matrix Summary:

- **Total Cases:** 93 | **Passed:** 89 (95.70%) | **Partially Verified:** 3 (3.23%) | **Platform Limitations:** 1 (1.08%) | **Failed:** 0 (0.00%)
- **TC-NOTIF-009 (Limitation):** Android Doze mode background delivery.
- **TC-CAL-005 (Partial):** Dynamic calendar date advancement boundary.
- **TC-AND-004 (Partial):** Physical LAN Wi-Fi API connectivity required host IP binding.
- **TC-AND-005 (Partial):** Sound verified; vibration not observed on physical Android 14 hardware.

25. IMPLEMENTATION VS. DOCUMENTATION GAP ANALYSIS

Discrepancies between common academic report templates and actual code:

Report / Theoretical Concept	Code Status	Actual Implementation Reality	Mandatory Report Correction
System Administrator Actor	NOT IMPLEMENTED	Only Patient and Caregiver roles exist in UserRole enum.	Remove Admin actor, admin use cases, and web portal diagrams.
Firebase Cloud Messaging (FCM)	NOT IMPLEMENTED	Reminders are 100% device-side local exact alarms via AlarmManager.	Replace 'Push Notifications' with 'Local AlarmManager Reminders'.
Prescription Photo OCR	NOT IMPLEMENTED	Medicine data is entered textually via AddMedicineScreen form.	Remove OCR camera capture claims and image processing pipelines.
AWS S3 / Cloud Storage	NOT IMPLEMENTED	PDF reports are generated dynamically in-memory (io.BytesIO).	Remove S3 bucket descriptions; describe in-memory streaming.
Relational Tables / SQL	NOT IMPLEMENTED	Database is MongoDB Atlas document store with 8 collections.	Represent ERD as logical document schema; explain ObjectId links.
Third-Party Email SDK (SendGrid)	NOT IMPLEMENTED	Standard library smtplib with TLS/SSL is implemented.	Document smtplib/MIME architecture; do not claim SendGrid/SES.

CRITICAL FACTS FOR REPORT COMPLETION

The following **30 authoritative facts** must be respected by any AI or human author completing academic chapters:

- **Fact 01:** Project official title is 'MediTrack+: Smart Medication Management & Adherence Platform'.
- **Fact 02:** Target actors are strictly Patient and Caregiver; Admin is NOT implemented.
- **Fact 03:** Mobile framework is React Native 0.86.3 with Expo SDK 57.0.21 and React 19.2.3.
- **Fact 04:** Backend framework is FastAPI 0.141.1 on Python 3.13 with Uvicorn ASGI on port 8000.
- **Fact 05:** Database is MongoDB Atlas using official PyMongo 4.18.0 Async API; Motor was replaced.
- **Fact 06:** There are exactly 8 MongoDB collections: users, medicines, medication_schedules, medication_activities, caregivers, notifications, scheduled_report_deliveries, rate_limits.
- **Fact 07:** There are exactly 38 registered REST endpoints across 9 routers.
- **Fact 08:** Reminders are 100% device-side local notifications; NO push server (FCM/APNs) is used.
- **Fact 09:** Android exact alarms use custom native Kotlin module (exact-alarm) querying AlarmManager.
- **Fact 10:** Android notification channel is 'medication-reminders' with MAX importance and ALARM audio stream.
- **Fact 11:** Password hashing uses Bcrypt with 12 computational rounds.
- **Fact 12:** Registration password min length is 6; password reset min length is 8.
- **Fact 13:** Authentication tokens are HS256 signed JWTs with 7-day expiration.
- **Fact 14:** Mobile JWT tokens are persisted in hardware-backed Expo SecureStore.
- **Fact 15:** Skip dose action requires a mandatory clinical skip reason (min length 1).
- **Fact 16:** Adherence percentage formula is (Taken Doses / Expected Scheduled Doses) * 100.0%.
- **Fact 17:** Adherence streak evaluates strictly consecutive adherent days backwards from today.
- **Fact 18:** Caregiver missed-dose alert Rule A requires >=2 missed doses in a rolling 24-hour window.
- **Fact 19:** Caregiver missed-dose alert Rule B requires 2 consecutive missed doses of the same medicine.
- **Fact 20:** Caregiver alerts are deduplicated via SHA-256 fingerprint in notifications collection.
- **Fact 21:** Caregivers have read-only clinical access; they CANNOT create/edit/delete patient medicines.
- **Fact 22:** PDF reports are rendered in-memory via ReportLab 5.0.1 and streamed directly over HTTP.
- **Fact 23:** Zero files are uploaded to AWS S3, Cloudinary, or remote cloud object storage.
- **Fact 24:** Email delivery supports MockEmailService and SMTPEmailService via standard smtplib.
- **Fact 25:** Prescription photo scanning / OCR camera upload is NOT implemented.
- **Fact 26:** There is no Admin web dashboard or admin portal.
- **Fact 27:** Automated pytest suite contains 385 tests across 24 test files.
- **Fact 28:** Formal test matrix contains exactly 93 test cases: 89 Passed, 3 Partial, 1 Limitation, 0 Failed.
- **Fact 29:** Physical Android 14 test verified visual banners and audio sound, but vibration was not observed.
- **Fact 30:** MediTrack+ is strictly a reminder/tracking tool; it contains authoritative medical safety disclaimers.

END OF AUTHORITATIVE PROJECT AUDIT & SOURCE OF TRUTH